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Visualisation, Data and Tasks

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SET09123 – Interactive Data Visualisation

Lecture Summary

- Data Visualisation
- Data & Semantics
- User Tasks

Data Visualisation

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What is your experience with Data Visualisation?

How would you define Data Visualisation?

A Definition

“The representation of information in the form of a chart, diagram, picture, etc.”

Oxford Dictionary

A Definition

“The representation of information in the form of a chart, diagram, picture, etc.”

Oxford Dictionary

*“The **purpose** of visualization is **insight**, not pictures.”*

Ben Shneiderman

Another Definition

*"**Computer**-based visualisation systems provide **visual representations** of **datasets** designed to help **people** carry out **tasks** more effectively."*

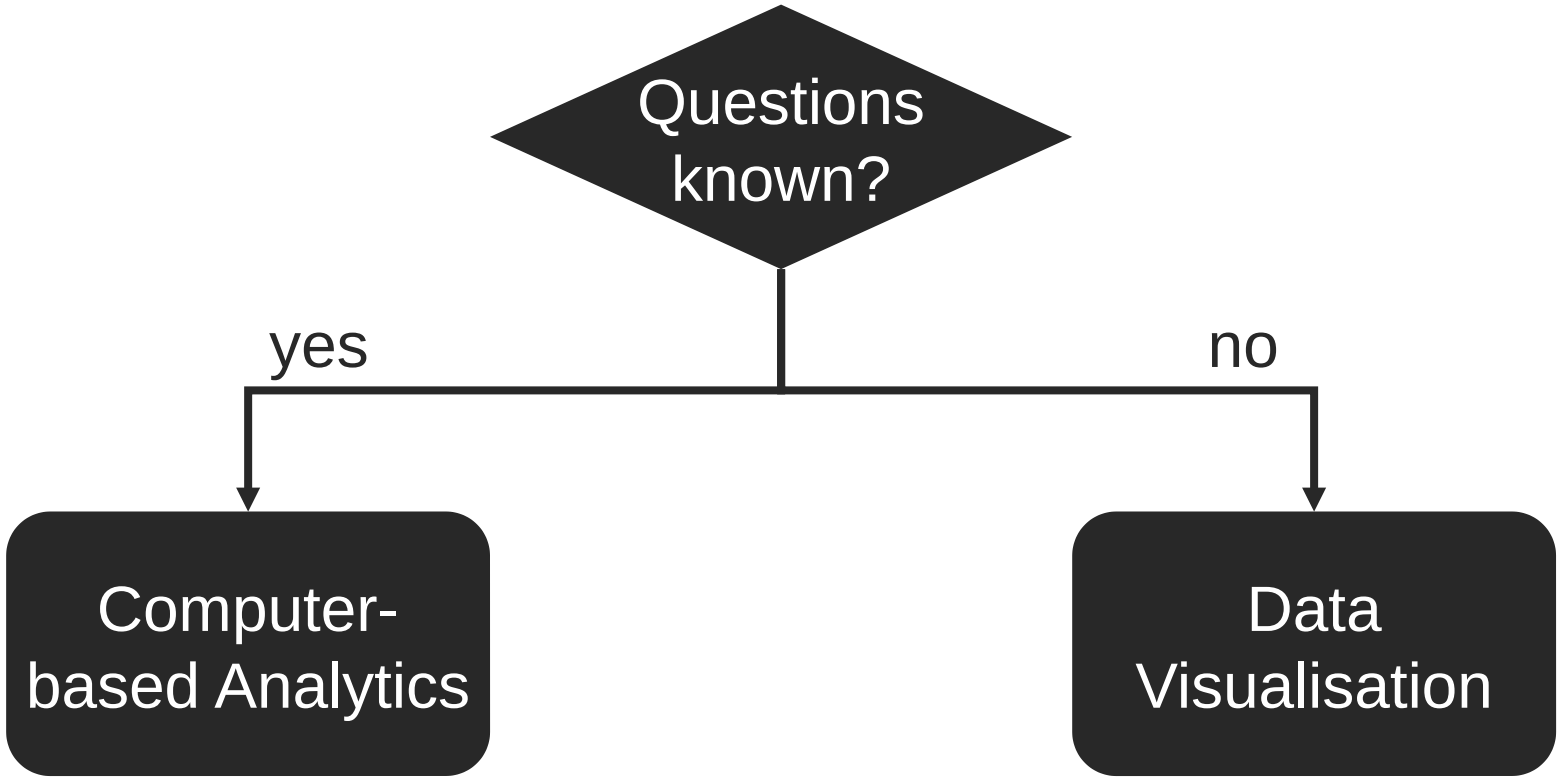
Tamara Munzner

An Analysis Cooperation Tool

- Data Visualisation enables the **analysis** of data

- It relies on the **cooperation** between **humans** and **machines**
  - A machine can compute and draw large datasets
  - A human can make decisions about how to analyse and interpret the patterns in data

An Analysis Cooperation Tool



Why Visualisation?

- To enable human analysis, we must present the whole data
  - Statistical summaries tend to loose details and important patterns
  - Example: *Anscombe Quartet*

Anscombe Quartet

Anscombe's Quartet: Raw Data

	1		2		3		4	
	X	Y	X	Y	X	Y	X	Y
	10.0	8.04	10.0	9.14	10.0	7.46	8.0	6.58
	8.0	6.95	8.0	8.14	8.0	6.77	8.0	5.76
	13.0	7.58	13.0	8.74	13.0	12.74	8.0	7.71
	9.0	8.81	9.0	8.77	9.0	7.11	8.0	8.84
	11.0	8.33	11.0	9.26	11.0	7.81	8.0	8.47
	14.0	9.96	14.0	8.10	14.0	8.84	8.0	7.04
	6.0	7.24	6.0	6.13	6.0	6.08	8.0	5.25
	4.0	4.26	4.0	3.10	4.0	5.39	19.0	12.50
	12.0	10.84	12.0	9.13	12.0	8.15	8.0	5.56
	7.0	4.82	7.0	7.26	7.0	6.42	8.0	7.91
	5.0	5.68	5.0	4.74	5.0	5.73	8.0	6.89
Mean	9.0	7.5	9.0	7.5	9.0	7.5	9.0	7.5
Variance	10.0	3.75	10.0	3.75	10.0	3.75	10.0	3.75
Correlation	0.816		0.816		0.816		0.816	

Anscombe, F. J. (1973). Graphs in statistical analysis. *The american statistician*, *27*(1), 17-21.
Images from Munzner (2014) (CC-BY-4.0)

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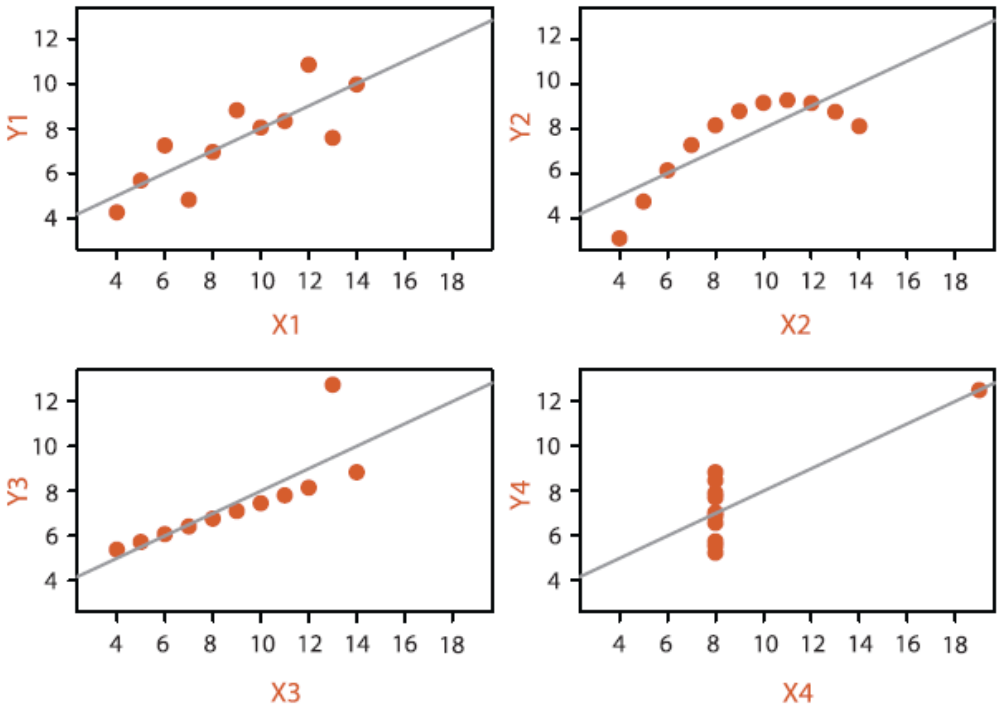

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Images from Munzner (2014) (CC-BY-4.0)

Why Visualisation?

- To facilitate human analysis, we must preserve cognitive efforts
  - An *external representation* allows to offload memory usage

- To maximise perception, we should use vision
  - Other channels such as touch, smell and taste have technological limitations
  - Sonification exists, but distinguishing sounds is less accurate
  - Vision is a high-bandwidth channel, with significant processing capabilities
  - There are accessibility concerns (e.g. Kim et al., 2021)

Kim, N. W., Joyner, S. C., Riegelhuth, A., & Kim, Y. (2021). Accessible visualization: Design space, opportunities, and challenges. In *Computer Graphics Forum* (Vol. 40, No. 3, pp. 173-188).

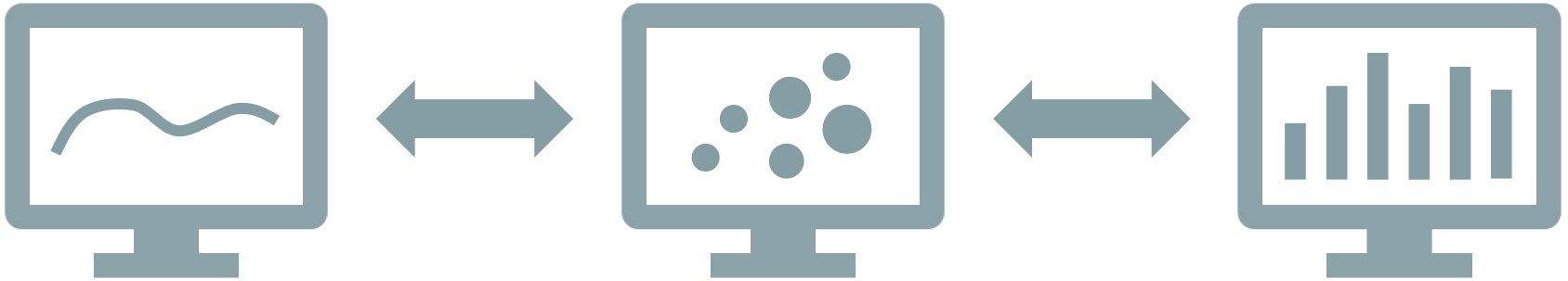
Why Interactivity?

- Humans, computer displays and static views have limited capabilities
  - Memory and attention (humans)
  - Screen resolution (displays)

- Large datasets cannot be presented so easily

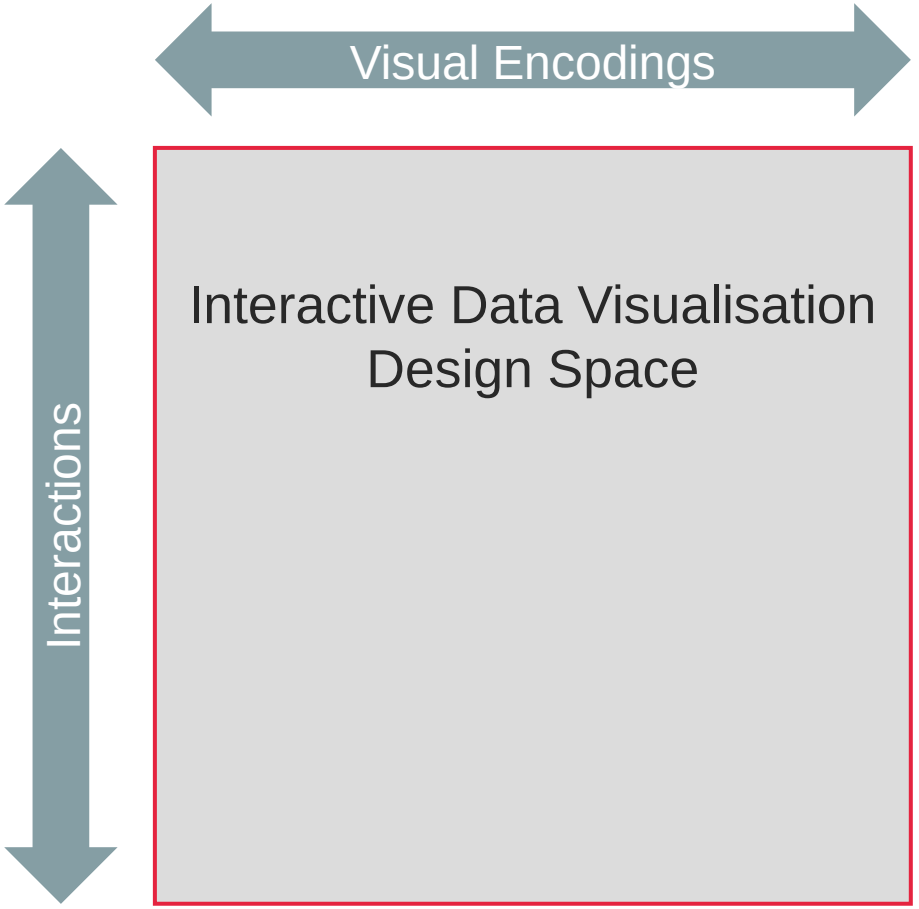
Why Interactivity?

- Interactivity enables the user to *adapt* the view to their analysis' needs:
  - Query different data subsets
  - Change between levels of abstraction
  - Find connections between parts of the dataset



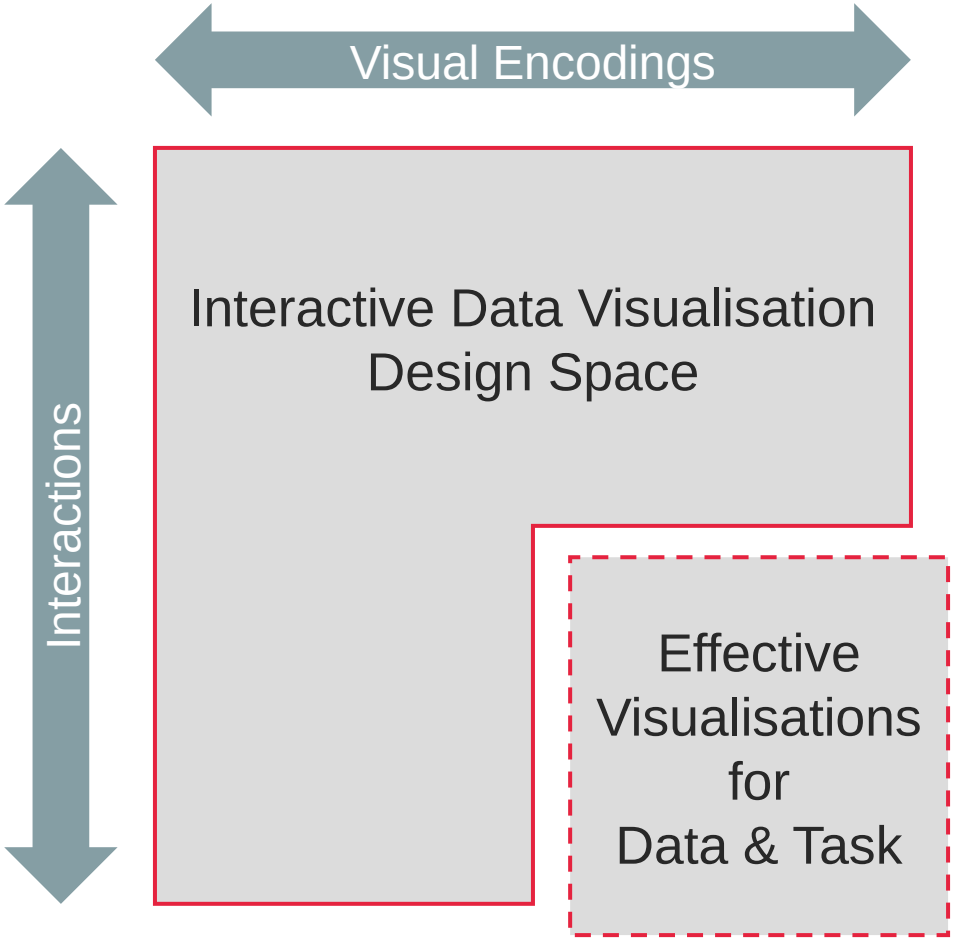
Design Space

- There are many **Visual Encodings** of data possible
- Adding **Interaction**, this makes a huge Design Space



Design Space

- ***How*** we construct interactive data visualisations needs to be analysed with:
  - ***what*** is shown (data) and
  - ***why*** the visualisation is needed (task)



Data & Semantics

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What type of data do you know of?

What semantics can you attach to it?

Data and Semantics

- Understanding *what* is the data we are trying to represent will limit the possibilities for visualisation

- The type of data and its semantics drive our choice for its visual arrangement

Positioning points based on geographical coordinates

vs.

Sizing areas based on categorical proportions

Types of Data

- We can define types at several levels
  - Attribute level
  - Data level
  - Dataset level

Attribute Types

They are typically 4 types of attribute

Nominal – Qualitative

- Categorical: categories
- Ordinal: ordered categories

Numerical – Quantitative

- Interval: numbers
- Ratio: numbers with a *true zero*

Attribute Types

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Focusing on *visualisation design*, we can differentiate these attribute types:

- Categorical: different *categories* with *no particular order*
- Ordered:
  - Ordinal: *ordered categories*, with no support for arithmetic
  - Numerical: measures that *support arithmetic*
  - Order types: sequential, diverging, cyclic

Data Types

- Attribute: measured property
- Item: individual entity
  - An item is characterised by a set of attributes
- Link: relation between items
  - Can be directed and undirected

- For *spatial* visualisation
  - Grid: sampling strategy in continuous space
  - Position: explicit location in 2D or 3D space

Dataset Types

- Tables: items as rows, attributes as columns
  - Multidimensional table
- Networks (graphs) & Trees: items (nodes) connected by links (edges)
  - Both nodes and links can have attributes
- Spatial visualisations:
  - Fields (continuous domain)
  - Geometry

Semantics

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Types of attribute and data types will drive design decisions…

… but so will semantics!

- Semantics is the **meaning** we attach to attributes and data

Semantics

- An attribute can be a **key** or a **value**
  - Keys are used to uniquely identify an item
    - Multidimensional data: 2 or more keys are necessary
  - Keys can be made of several attributes

Semantics

- Categorical attributes can have a hierarchical relationship
- A quantitative attribute can represent a measure, a spatial property or a temporal feature
- A link can represent similarity, hierarchy or dependency

User Tasks

Why do you use Data Visualisation?

User Tasks

- Thinking about *why* a user needs an interactive visualisation drastically changes the way we design it

- A tool designed and effective for one task, may not be suitable for another

Comparing correlations between multiple dimensions

vs.

Understanding the relative proportion of one category in the whole dataset

Actions and Targets

- There are two aspects to tasks:

Actions: what the user does

e.g., *compare, explore, present*

Targets: which aspect of the data interests the user

e.g., *trends, similarity, features*

Action: Analyse

To **inspect** a visualisation in order to **interpret** and **explain** the data

- *Discover*: find new knowledge
- *Present*: communicate information

- *Annotate*: add new (temporary) information
- *Record*: extract (persistent) artifacts
- *Derive*: create new, transformed, data

Action: Search

To **find** elements of **interest** (targets) in the visualisation

- *Lookup*: target and location known
- *Locate*: target know but location unknown
- *Browse*: target unknown but location known
- *Explore*: target and location unknown

Action: Query

Once found, **examine** the targets to:

- *Identify* the attributes of a single target
- *Compare* attributes between two targets
- *Summarise* the attributes of all targets

Targets: Dataset

Targets can be over the whole dataset, on a high-level

- Trends or patterns in the dataset
- Outlier data points
- Domain-specific features

Targets: Attribute

Targets can be specific attributes

- Distribution and extremes of one attribute
- Correlation and similarity between multiple attributes
- Data-specific structure:
  - Network topology
  - Spatial shapes

Conclusion

Key Points

- Data Visualisation enables the **cooperation** between humans and computers to **analyse** datasets
- Interactivity bypasses limitations and facilitates the analysis of large dataset
- Designing visualisations and interactions needs to take into account:
  - The data: types of attributes, data and dataset and their semantics
  - The user tasks: why a user might want to interact with the visualisation